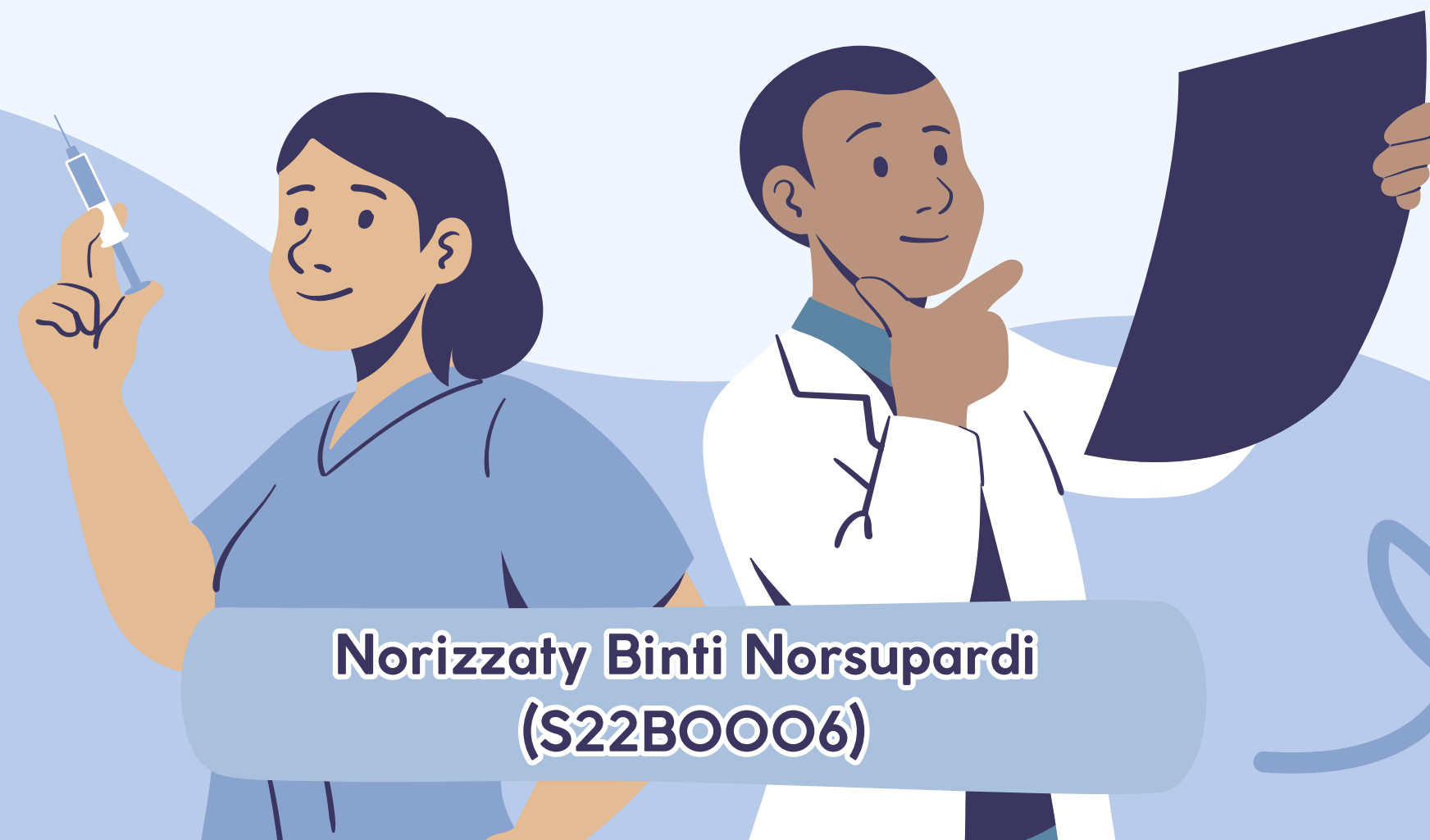


Heart Failure

Detection using SVM and LR



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(S22B0006)

Problem Framing & Data Exploration

Objectives

Using classification for predicting the likelihood of heart disease.

Dataset

- Heart Failure Prediction dataset
- Kaggle
- 918 patients
- Target: Heart Disease (1 = present, 0 = absent)
- Features: Categorical & Numerical

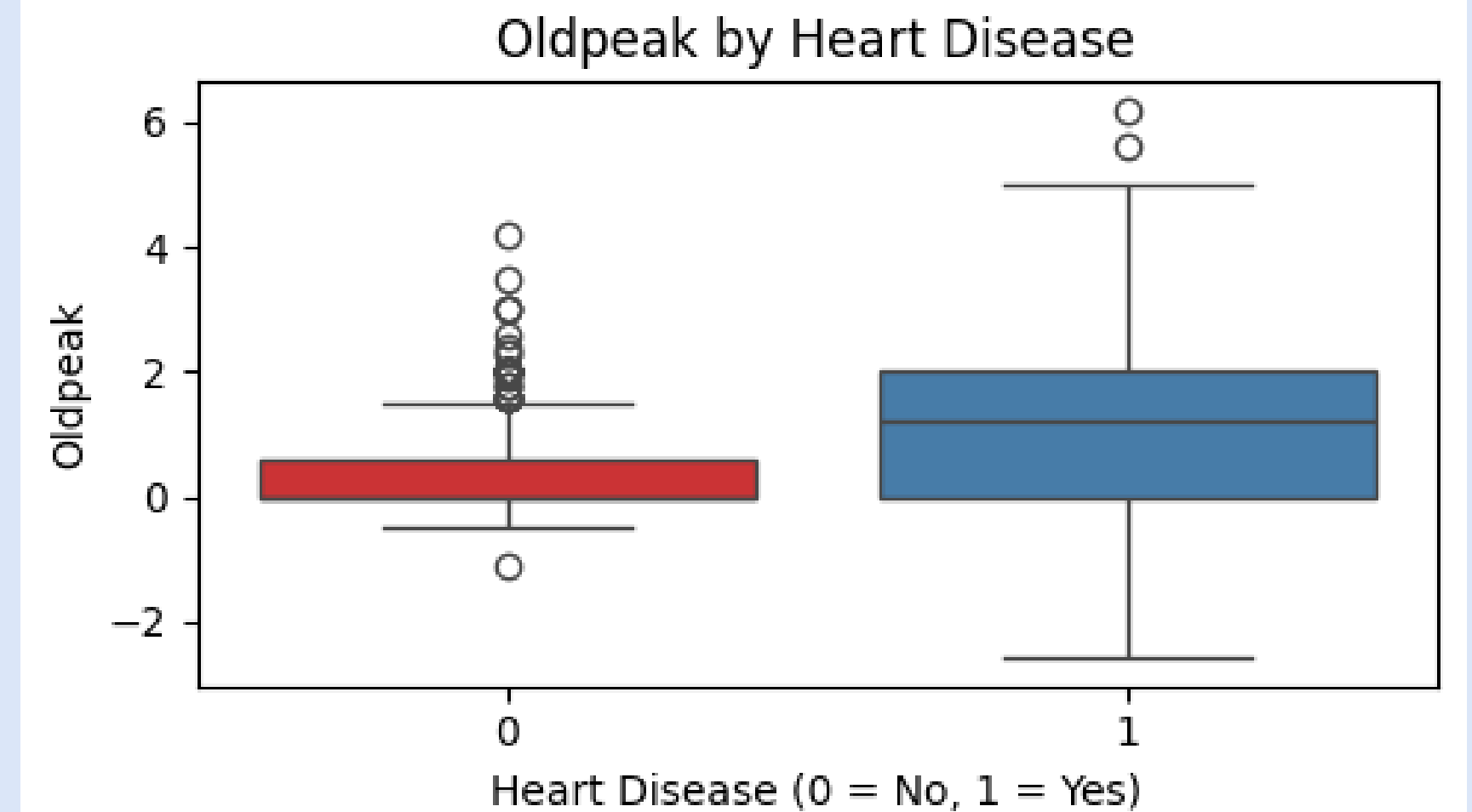
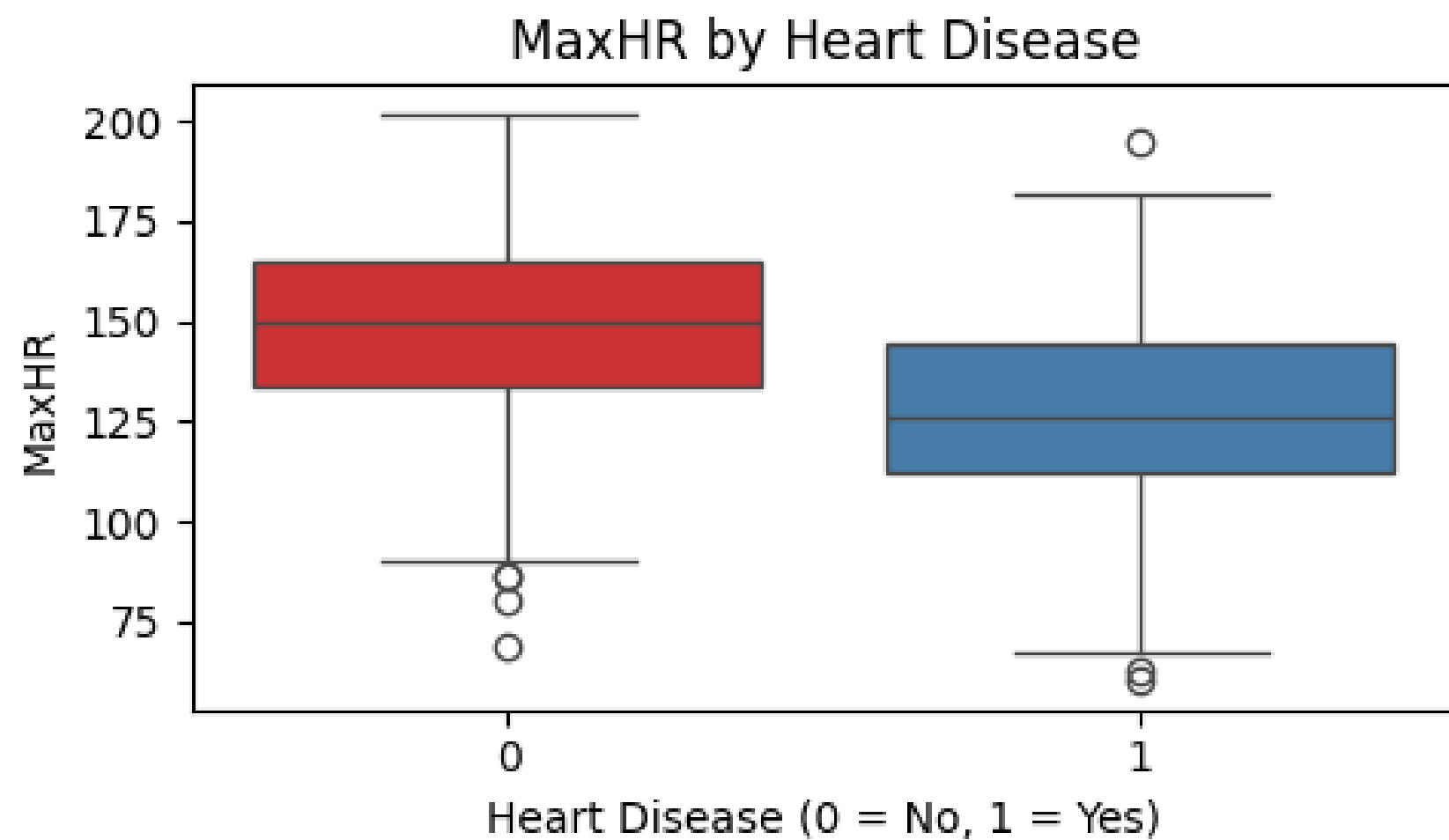
Exploration Insight

Patients with heart disease often show:

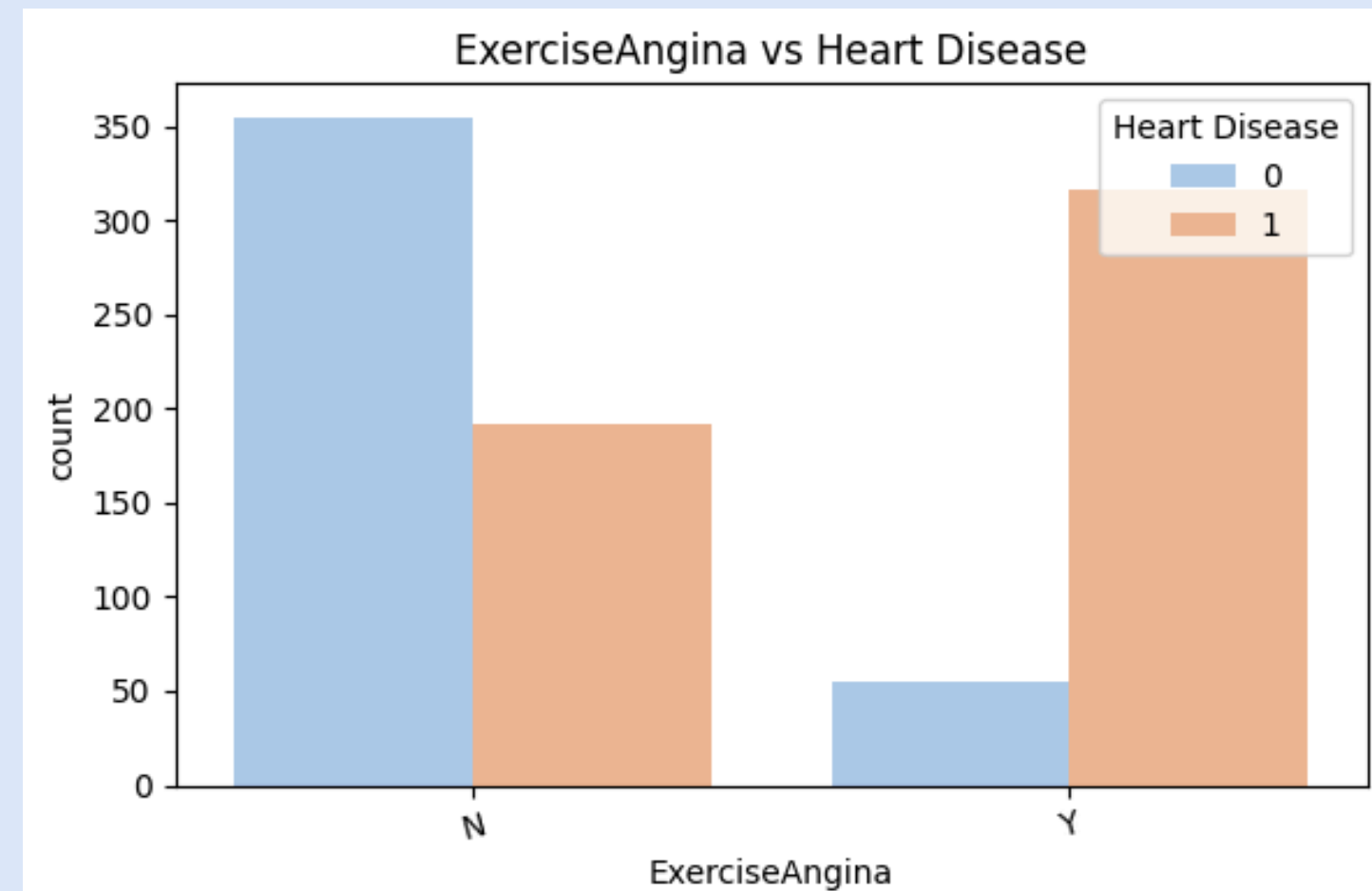
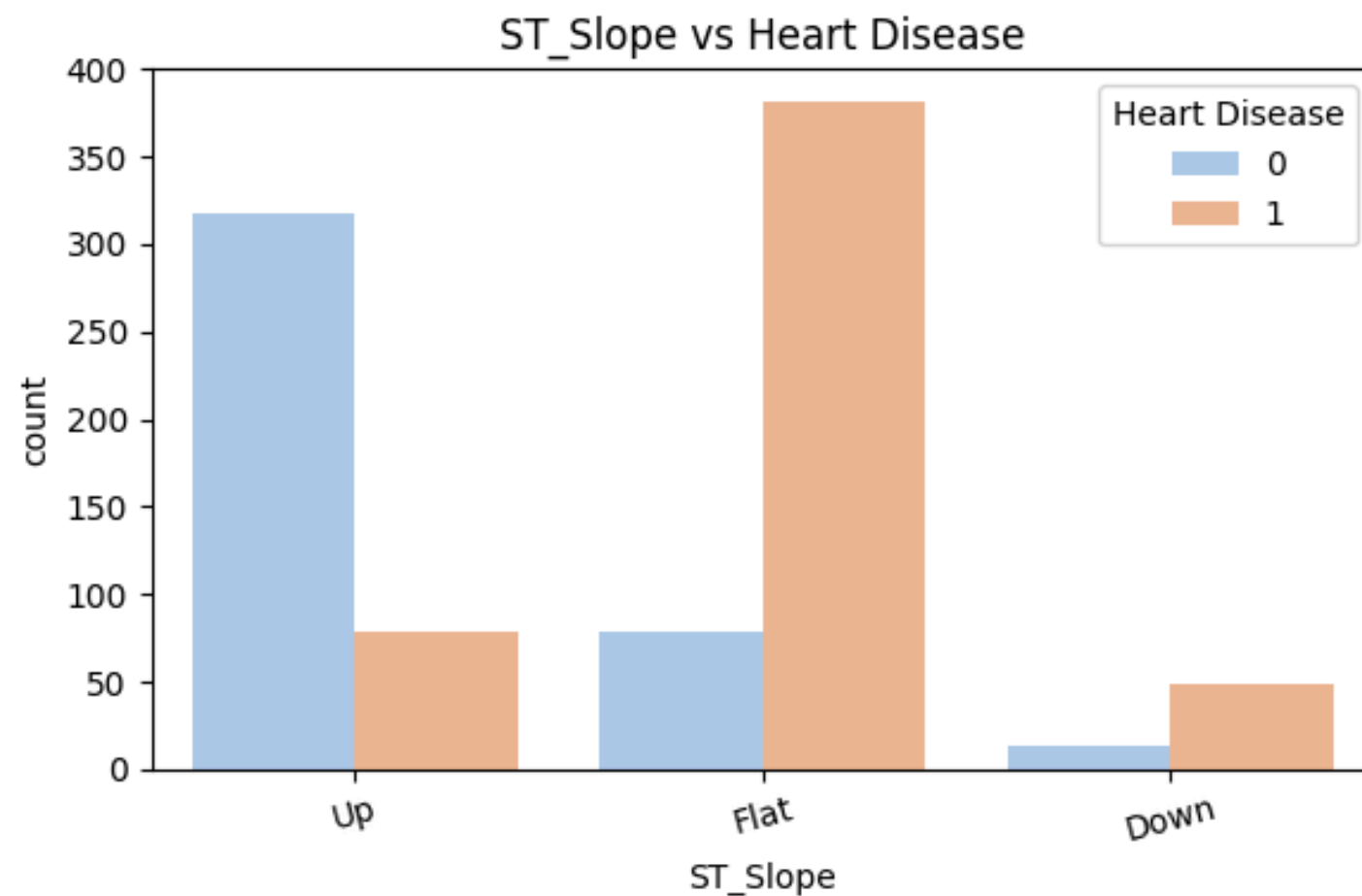
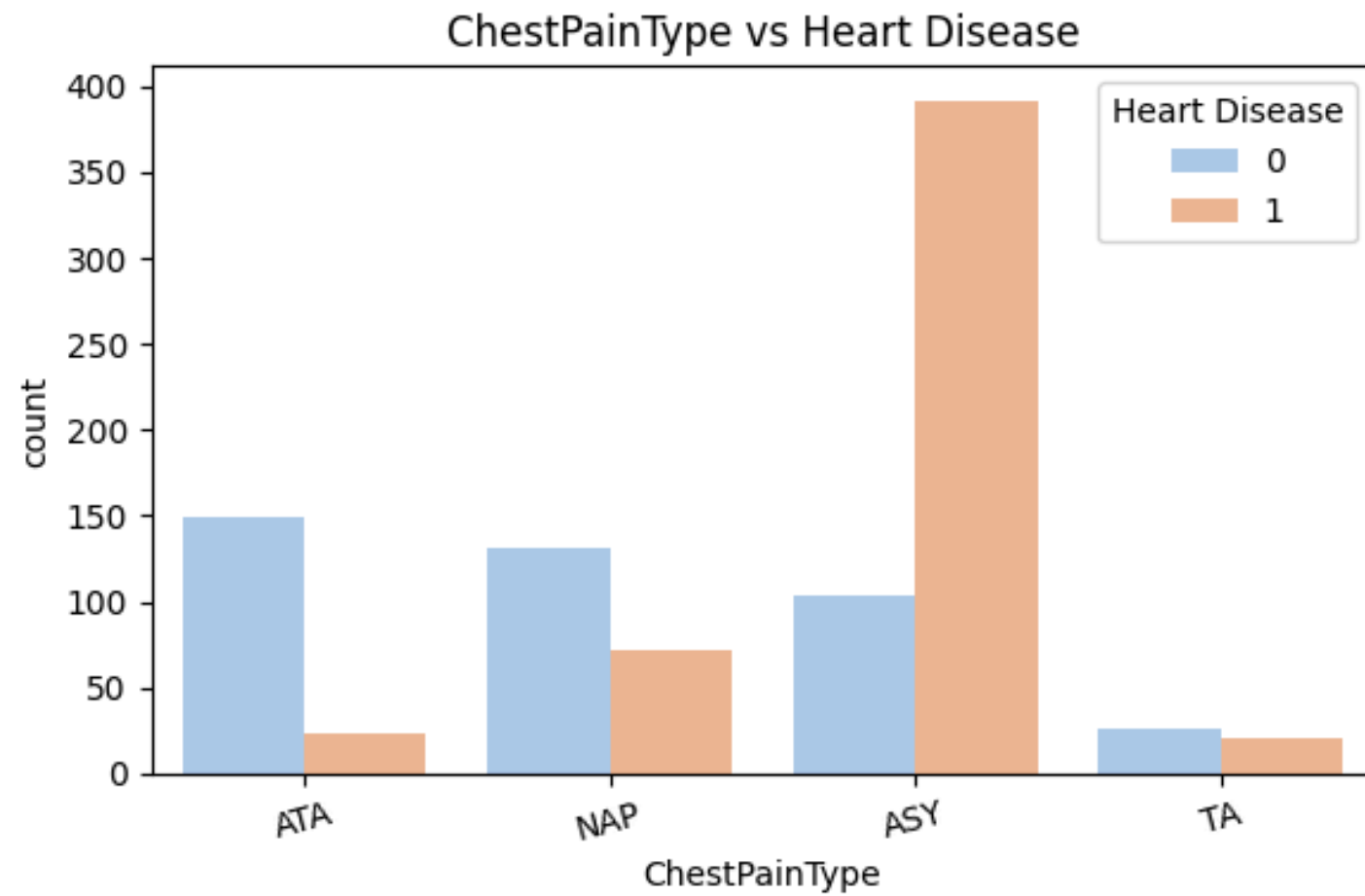
- Lower MaxHR
- Certain ChestPainType patterns
- Differences in ST_Slope and Exercise Angina



Boxplot



Bar Chart



Model Implementation

Model Used:

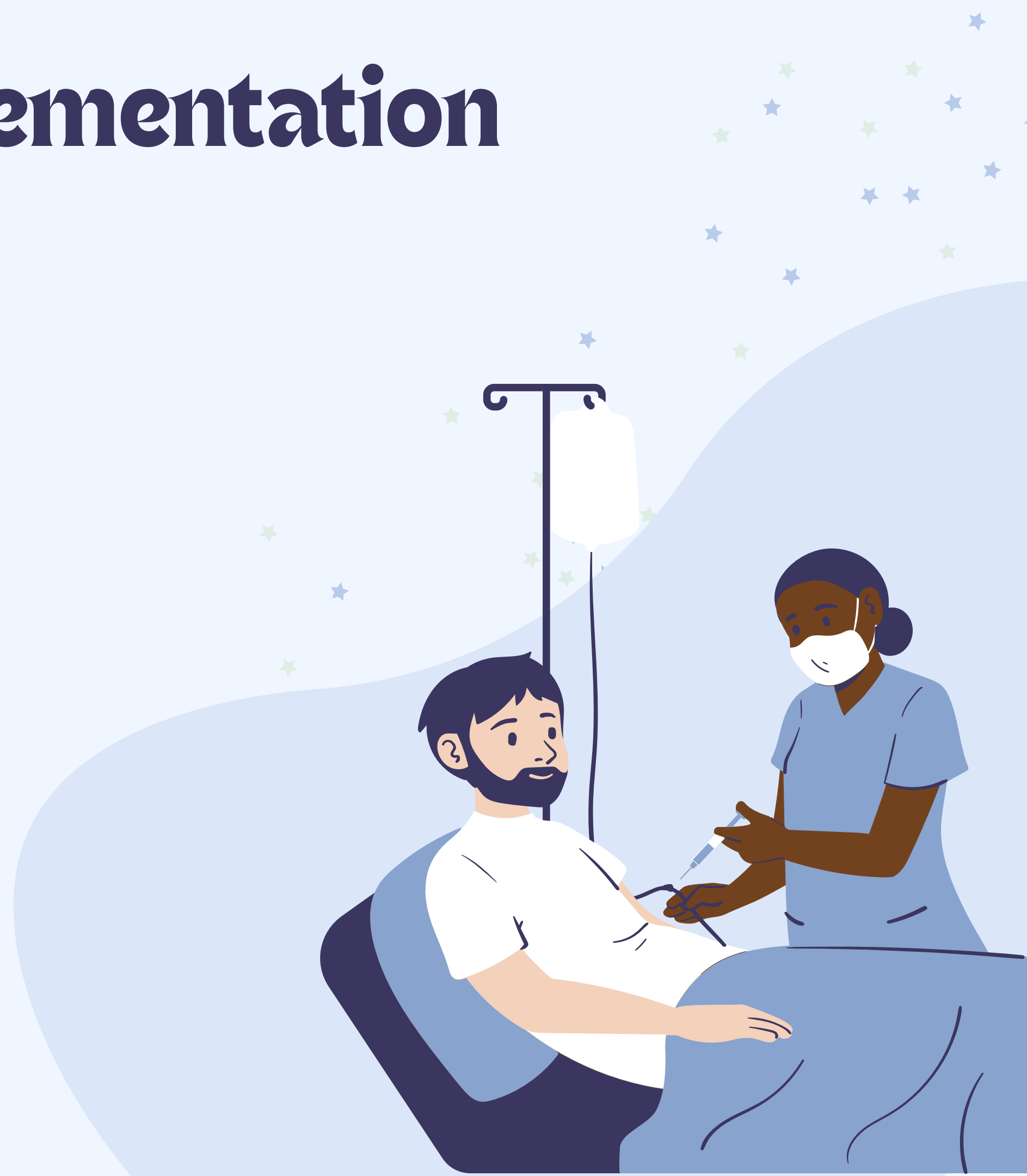
- Support Vector Machine (SVM)
- Logistic Regression (LR)

Why These Models?

- SVM: Good at finding optimal boundaries in high dimensional space
- LR: Easy to interpret and provides probabilistic insights

Software & Libraries

- Python in VS code
- Libraries: pandas, scikit-learn, matplotlib, seaborn



Model Building & Training

Preprocessing Steps:

- Removed Duplicates
- Categorical variables encoded using `pd.get_dummies`
- Standardized numerical features using `StandardScaler`

Feature Selection:

- Using all 11 features

Model Training:

- 80/20 Train test split
- SVM: `GridSearchCV` used for tuning kernel, C, gamma, degree
- LR: Applied with default settings and standardized input



Evaluation & Metric Comparison

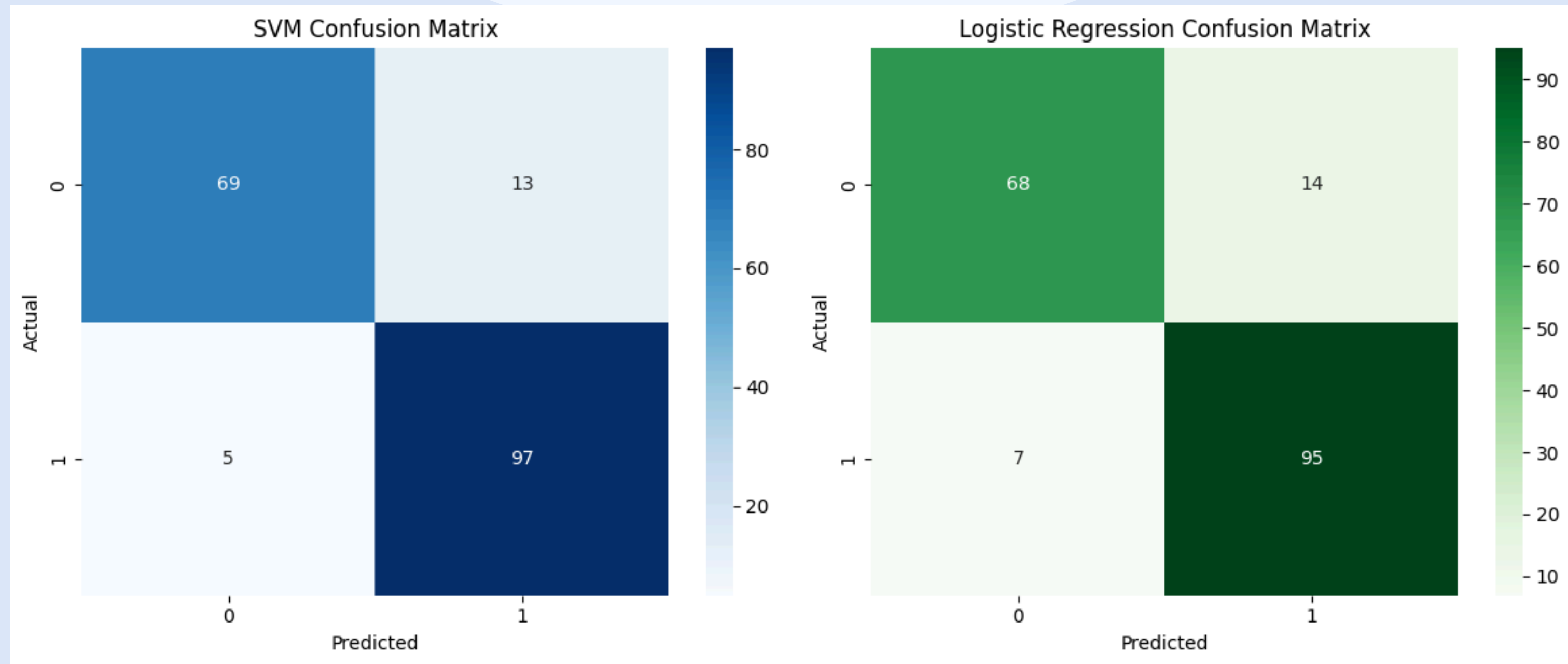
Classification Report for LR

Classification Report for SVM

	precision	recall	f1-score	support
No Disease (0)	0.91	0.83	0.87	82
Heart Disease (1)	0.87	0.93	0.90	102
accuracy			0.89	184
macro avg	0.89	0.88	0.88	184
weighted avg	0.89	0.89	0.89	184

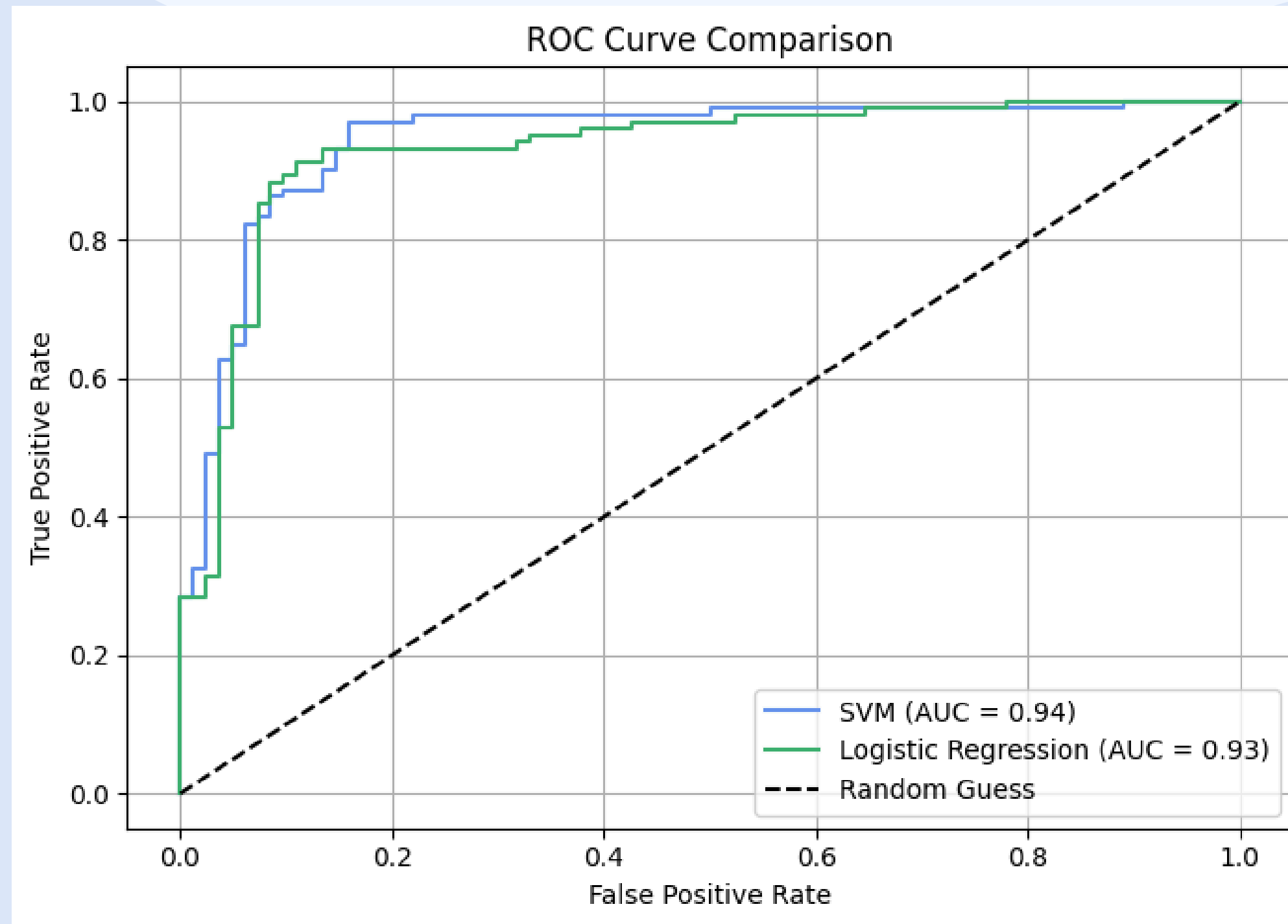
	precision	recall	f1-score	support
No Disease (0)	0.93	0.84	0.88	82
Heart Disease (1)	0.88	0.95	0.92	102
accuracy			0.90	184
macro avg	0.91	0.90	0.90	184
weighted avg	0.90	0.90	0.90	184

Evaluation & Metric Comparison



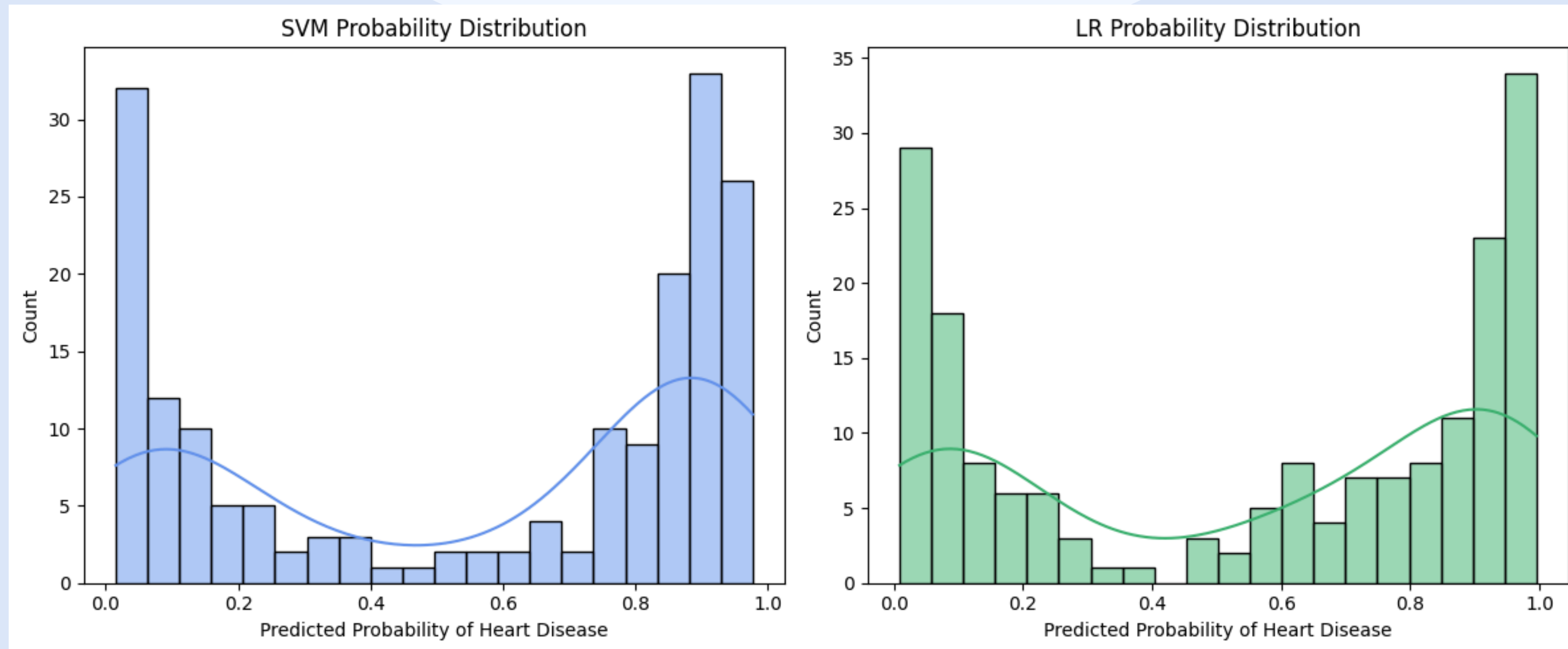
Side-by-side Heatmap for SVM and LR

Evaluation & Metric Comparison



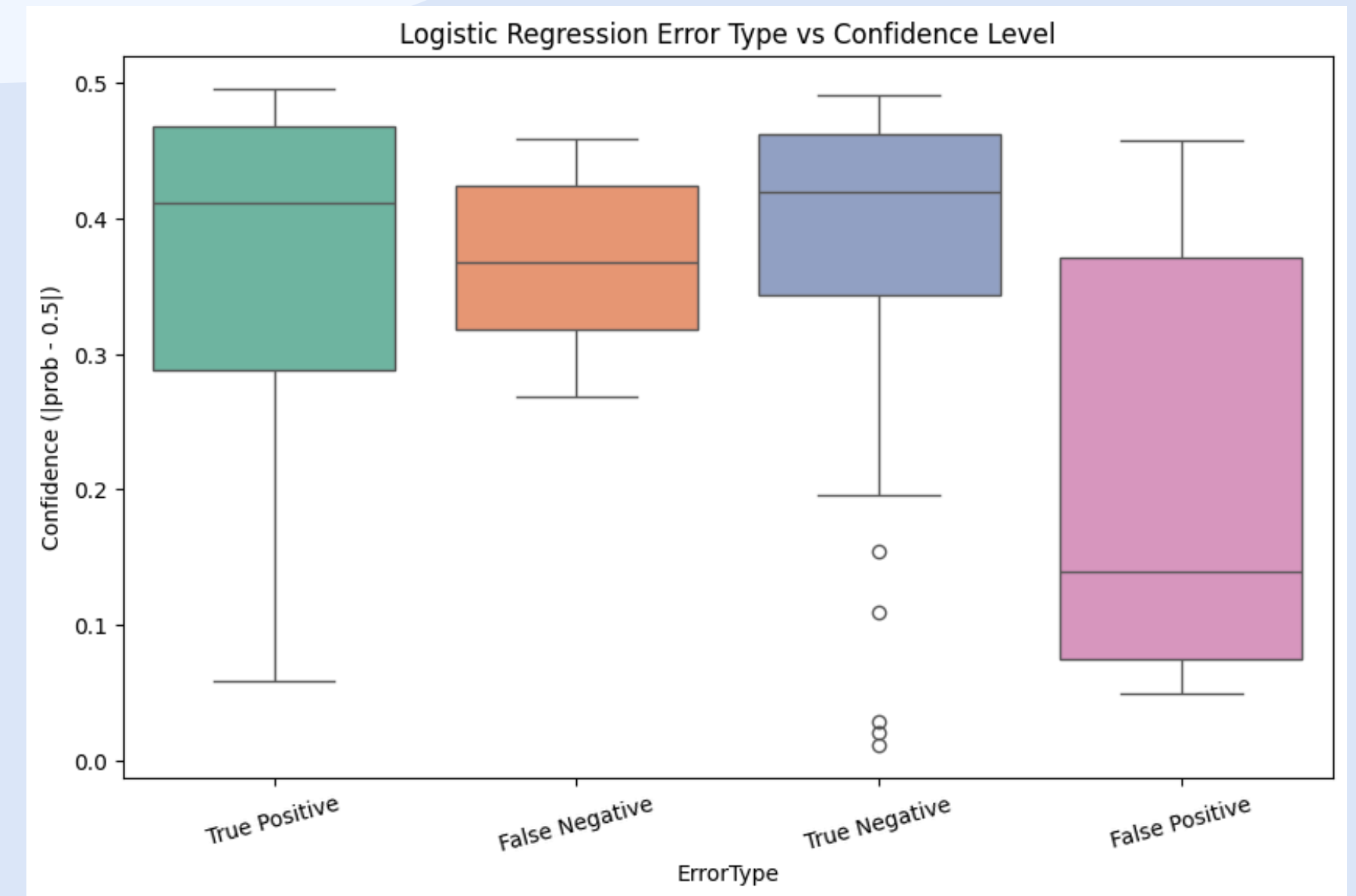
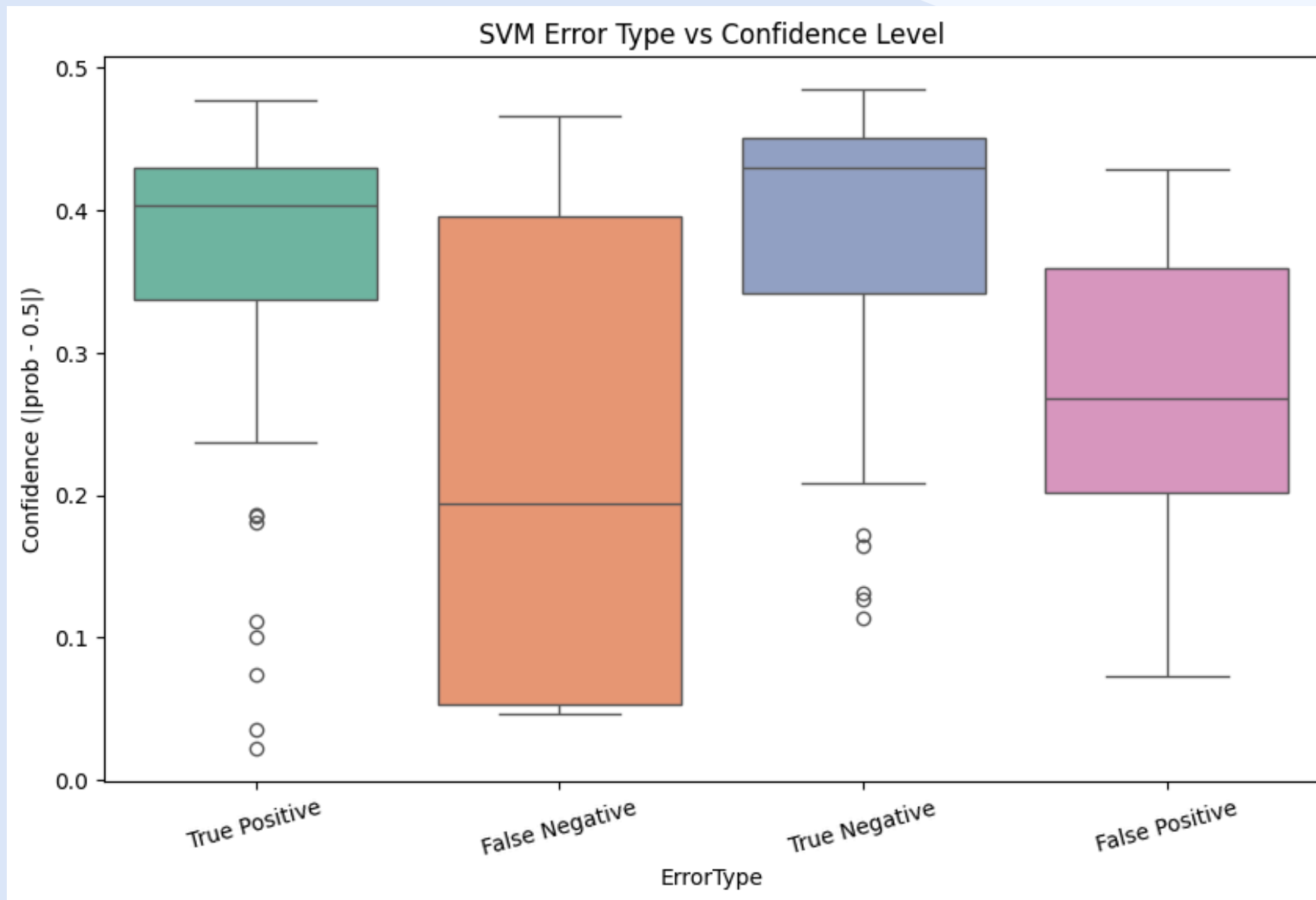
ROC Curve Comparison for SVM and LR

Evaluation & Metric Comparison



Histogram for SVM and LR

Evaluation & Metric Comparison



Box Plot for Error Type vs SVM and LR

Discussion

Model Comparison & Performance

- SVM consistently outperformed LR all across metric evaluation
- SVM handled complex relationship better
- LR remained interpretable and computationally faster

Experimentation & Improvements

- Tried different train-test splits (e.g. 70/30), resulting slightly lower accuracy.
- Dropping certain features 'Age' and 'Sex' but performance was decreased



Discussion

Unexpected Finding:

- LR occasionally produced more false positive, indicates slight bias in predicting disease.

Limitation:

- Class imbalance slightly affected prediction confidence
- Feature limitation: no lifestyle/genetic factor included

Recommendation:

- Expand Dataset
- Try Ensemble Method (e.g. Random Forest, Voting Classifier)



Conclusion

Chosen based on strengths:

- SVM: high performance
- LR: Interpretability

Consistent results through:

- Data preprocessing
- Feature selection
- Model tuning



References

1. SVM model

Heart Failure Detection SVM

2. LR model

Heart Failure detection LR

3. SVM + LR results comparison

Model Comparison side by side



**Thank you for
your attention**

